

MATH 301: Homework 5

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Section 3.3 : 2 Let $x_1 > 1$ and $x_{n+1} := 2 - 1/x_n$ for $n \in \mathbb{N}$. show that (x_n) is bounded and monotone. Find the limit.

Monotone Sequence Let $X = (x_n)$ be a sequence.

- X is increasing if $x_1 \leq x_2 \leq \dots \leq x_n \leq \dots$
- X is decreasing if $x_1 \geq x_2 \geq \dots \geq x_n \geq \dots$
- X is monotone if it is either increasing or decreasing.

Note the following relationship:

$$1 < x_1 \implies \frac{1}{x_1} < 1 \quad \text{Let } \lambda_n := \frac{1}{x_n}$$

Observe first few values of X :

$$x_2 = 2 - \lambda_1 \quad x_3 = 2 - \lambda_2 \quad \dots \quad x_n = 2 - \lambda_{n-1}$$

The nature of the growth/decay of the function is dependent on the values of the λ term.

Notice (Subproblem A1):

$$\begin{aligned} &\text{Observe } x_1 \in (1, \infty) \text{ Suppose } x_k \in (1, \infty) \\ \implies &1 < x_k \implies \frac{1}{x_k} < 1 \implies 0 < 1 - \frac{1}{x_k} \\ \implies &1 < 2 - \frac{1}{x_k} \equiv 1 < x_{k+1} \\ \therefore &\forall n \in \mathbb{N}, \text{ by PMI } x_n \in (1, \infty) \end{aligned}$$

$$\text{Notice: } x_{k+1} - x_k := \left(2 - \frac{1}{x_k}\right) - x_k = \frac{x_k - 2x_k + 1 - x_k^2}{x_k} = -\frac{(x_k - 1)^2}{x_k}$$

Note from Subproblem A1 we can construct these three statements:

$$0 < x_n \quad 0 < (x_n - 1)^2 \quad -\frac{(x_k - 1)^2}{x_k} < 0$$

As $x_{k+1} - x_k < 0$. This implies $x_{k+1} < x_k$. Thus the sequence is monotonic and decreasing. Therefore a limit must exist.

Recall the limit of the tail of any sequence must converge to the same limit of the sequence as for any sequence we are guaranteed the uniqueness of any limit.

$$\begin{aligned} \implies \lim_{n \rightarrow \infty} x_n = L &\implies \lim_{n \rightarrow \infty} x_{n+1} = L \implies L = 2 - \frac{1}{L} \\ &\therefore L = 1 \end{aligned}$$

Section 3.3 : 3 Let $x_1 \geq 2$ and $x_{n+1} := 1 + \sqrt{x_n - 1}$ for $n \in \mathbb{N}$. Show that (x_n) is decreasing and bounded below by 2. Find the limit.

$$\begin{aligned} & \text{Observe } 2 \leq x_1 \quad \text{Suppose } 2 \leq x_k \\ & \left\{ \begin{array}{ll} 2 = x_k & \implies x_{k+1} = 1 + \sqrt{2-1} = 2 = x'' \\ 2 < x_k & \implies x_{k+1} = 1 + \sqrt{x_k - 1} = 1 + \sqrt{x_1} = x' \end{array} \right. \implies \left\{ \begin{array}{ll} 2 = x_k & \implies 2 = x_{k+1} \\ 2 < x_k & \implies 2 < x_{k+1} \end{array} \right. \\ & \therefore \forall n \in \mathbb{N}, 2 \leq x_n \end{aligned}$$

Therefore we have shown there is a lower bound to the sequence.

$$\begin{aligned} & \text{Observe: } 2 \leq x_1 \implies 1 \leq x_1 - 1 \implies 1 \leq \sqrt{x_1 - 1} \implies 2 \leq 1 + \sqrt{x_1 - 1} \\ & \text{Suppose } x_{k+1} \leq x_k \\ & \text{Let } \begin{cases} x_{k+2} := 1 + \sqrt{x_{k+1} - 1} \\ x_{k+1} := 1 + \sqrt{x_k - 1} \end{cases} \\ & x_{k+1} < x_k \implies \sqrt{x_{k+1} - 1} < \sqrt{x_k - 1} \implies x_{k+2} \leq x_{k+1} \\ & \therefore \forall n \in \mathbb{N}, x_n \leq x_{n-1} \end{aligned}$$

Therefore we have shown the sequence is decreasing.

Consider:

$$\begin{aligned} \lim x_n = x & \implies 1 + \sqrt{x - 1} = 0 \\ \lim x_n & = 1 \vee \lim x_n = 2 \end{aligned}$$

As x_1 is defined to be $2 \leq x_1$ the limit must be the bound of the sequence 2.

Section 3.3 : 9 Let A be an infinite subset of $\mathbb{R}$ that is bounded above and let $u := \sup A$. Show there exists an increasing sequence (x_n) with $x_n \in A$ for all $n \in \mathbb{N}$ such that $u = \lim(x_n)$.

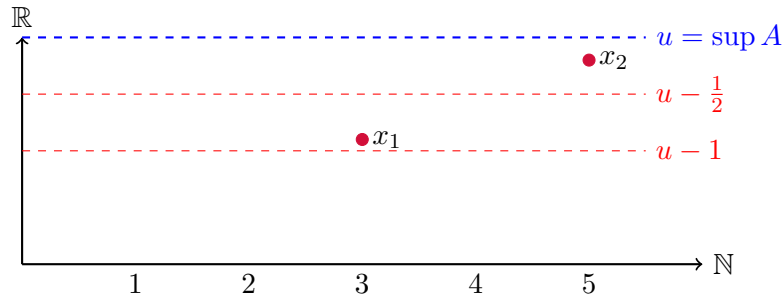
If $A \subseteq \mathbb{R}$ and $\sup A = u \implies A := \{a_1, a_2, \dots, a_n, a_{n+1}, \dots \mid a \in \mathbb{R}\}$ and $\forall a \in A, a \leq u$.

$$\sup A = u \implies \forall n \in \mathbb{N}, \exists a'_n \in A \text{ s.t. } u - \frac{1}{n} < a'_n < u$$

$$\text{Let } x_1 := u - 1 < a'_1 < u \quad \text{Let } x_2 := \max(x_1, a'_2)$$

$$\text{Generally, } x_n := \max(x_{n-1}, a'_n)$$

Visualization



We are defining a line $u - \frac{1}{n}$ and returning the smallest value of a that is larger than $u - \frac{1}{n}$ and assigning it to the corresponding n . This a is what we have called a'_n (The minimum value of a which depends on n that shows that $u - \epsilon$ is not an upper bound)

Consider we have constructed in a sequence defined to be : $x_{n-1} \leq x_n$. Thus the sequence is increasing.

Notice as $x_n := \max(x_{n-1}, a'_n)$. This implies $a'_n \leq x_n$. Thus $u - \frac{1}{n} < a'_n \implies u - \frac{1}{n} < x_n$

Therefore for all terms $u - \frac{1}{n} < x_n \leq u$

$$\lim_{n \rightarrow \infty} \left(u - \frac{1}{n} < x_n \leq u \right)$$

$$u \leq x_n \leq u \implies \lim_{n \rightarrow \infty} x_n = u$$

Section 3.3 : 11 Let $x_n := 1/1^2 + 1/2^2 + \dots + 1/n^2$ for each $n \in \mathbb{N}$. Prove that (x_n) is increasing and bounded, and hence converges. [*Hint*: Note that if $k \geq 2$, then $1/k^2 \leq 1/k(k-1) = 1/(k-1) - 1/k$.]

Consider

$$\text{Let } b_n := \frac{1}{n^2} \implies x_n := \sum_{n=1}^{\infty} b_n$$

$$\text{Observe: } b_1 < b_1 + b_2 \implies x_1 < x_2$$

$$\text{Suppose: } x_k < x_{k+1} \implies \sum_{n=1}^k b_n < \sum_{n=1}^{k+1} b_n$$

$$\implies b_{k+2} + \sum_{n=1}^k b_n < b_{k+2} + \sum_{n=1}^{k+1} b_n \implies x_{k+1} < x_{k+2}$$

$\therefore \forall n \in \mathbb{N}, x_{n-1} < x_n$ Therefore we can say the sequence is increasing

Note

$$S := \sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \sum_{n=2}^{\infty} \frac{1}{n^2}$$

$$\forall n \in \mathbb{N}, 2 \leq n \implies n(n-1) < n^2 \implies \frac{1}{n^2} < \frac{1}{n(n-1)}$$

Consider

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < 1 + \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right)$$

Observe (Canceling Terms)

$$\sum_{n=2}^k \left(\frac{1}{n-1} - \frac{1}{n} \right) = \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) \dots + \left(\frac{1}{k-1} - \frac{1}{k} \right) = 1 - \frac{1}{k}$$

We could also take the gaussian summation of the individual terms confirmed with induction and show it equals to the same value.

$$\lim_{k \rightarrow \infty} 1 - \frac{1}{k} = 1 \implies \sum_{n=1}^{\infty} \frac{1}{n^2} < 2$$

Therefore x_n is mononic, increasing, and bounded above, therefore there must converge.

Section 3.4 : Use method of *Example 3.4.3(b)* to show $0 < c < 1 \implies \lim c^{\frac{1}{n}} = 1$

Example (b): Proving $\lim(c^{1/n}) = 1$ for $c > 1$

The Goal: Prove that the n -th root of a number greater than 1 gets closer and closer to 1 as n gets larger. Let our sequence be $z_n = c^{1/n}$.

Step 1: Show the sequence is decreasing

The text asks "(Why?)" here. Here is the reason: Since $c > 1$, taking a smaller and smaller fractional power of c yields a smaller and smaller result. Because $\frac{1}{n+1} < \frac{1}{n}$, it follows that:

$$\begin{aligned}c^{1/(n+1)} &< c^{1/n} \\ z_{n+1} &< z_n\end{aligned}$$

So, the sequence is decreasing.

Step 2: Show the sequence is bounded

Because $c > 1$, any positive root of c will still be greater than 1.

$$z_n > 1$$

The sequence is bounded below by 1.

Step 3: Conclude it converges

Just like in example (a), we have a decreasing sequence that is bounded below. By the Monotone Convergence Theorem, it converges. Let's call the limit z :

$$z = \lim(z_n)$$

Step 4: Find the limit using the same trick

Let's look at the subsequence of even terms, z_{2n} . Because the main sequence converges to z , this subsequence also converges to z . Let's look at the algebra:

$$z_{2n} = c^{1/(2n)} = (c^{1/n})^{1/2} = \sqrt{z_n}$$

If we square both sides to get rid of the fraction, we get:

$$(z_{2n})^2 = z_n$$

Now, take the limit of both sides as $n \rightarrow \infty$:

$$z^2 = z$$

Step 5: Solve for z

Again, the only solutions to $z^2 = z$ are $z = 0$ and $z = 1$. Since we established in Step 2 that every term in our sequence is strictly greater than 1 ($z_n > 1$), the limit cannot possibly be 0. Therefore, the limit must be 1.

$$\lim(c^{1/n}) = 1$$

Section 3.4 : Use method of *Example 3.4.3(b)* to show $0 < c < 1 \implies \lim c^{\frac{1}{n}} = 1$

$$0 < c < 1 \implies \exists a < b, \in \mathbb{R} \text{ s.t. } c := \frac{a}{b}$$

Observe: $0 < c < 1 \implies \lim_{n \rightarrow \infty} c^n = 0 \implies \forall n \in \mathbb{N}, c_n \in (0, 1] \wedge c^n < c^{n+1}$

$$c^n \cdot c^{\frac{1}{n^2}} < c^{n+1} \cdot c^{\frac{1}{n^2}}$$

$$c^{\frac{1}{n}} < c^{\frac{1}{n} + \frac{1}{n^2}}$$

$$\therefore c^{\frac{1}{n}} < c^{\frac{1}{n+1}}$$

$\therefore c^{n+1} < c^n < c^{\frac{1}{n}} < c^{\frac{1}{n+1}} \implies c^n$ is monotonic and increasing.

Let c_{2n} be a subsequence of c_n containing even naturals

$$z_{2n} := \left(\frac{a}{b}\right)^{\frac{1}{2n}} \quad z_n := \left(\frac{a}{b}\right)^{\frac{1}{n}}$$

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